

YUDONG LI

M.Sc. Student | Robot Planning, Control, and Embodied Autonomy

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Research interests: robot trajectory planning and motion control; robot learning; embodied navigation and manipulation; Sim-to-Sim and Sim-to-Real deployment; and embedded robotic systems.

EDUCATION

Southern University of Science and Technology (SUSTech) | Shenzhen, China

M.Sc. in Intelligent Manufacturing and Robotics, Department of Mechanical and Energy Engineering Sep 2024 - Jun 2027 (expected)

- GPA: 3.08/4.00; Intelligent Robotics and Control Laboratory.

South China Agricultural University | Guangzhou, China

B.Eng. in Automation, College of Engineering Sep 2019 - Jun 2024

- GPA: 4.03/5.00; weighted average: 90.64/100; rank: 4/107; Excellent Undergraduate Thesis.

PUBLICATIONS AND PATENT

1. Guoyang Zhao*, Yudong Li*, Weiqing Qi, Kai Zhang, Bonan Liu, Kai Chen, Haoang Li, and Jun Ma. "Decision-Driven Semantic Object Exploration for Legged Robots via Confidence-Calibrated Perception and Topological Subgoal Selection." IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. Accepted. *Equal contribution. arXiv:2509.20739.
2. Yudong Li, Zehao Huang, Zenghang Huang, Guojian Feng, and Bolin Liang. "Design and Prototype Experiment of a Gimbal System for Agricultural Patrol Robots." Southern Agricultural Machinery, 54(22):24-27, 2023. First author.
3. Zekai Huang, Yudong Li, Zenghang Huang, Zehao Huang, et al. "Design and Experiment of an Adaptive Mechanical Chassis for Hilly Farmland." Southern Agricultural Machinery, 54(23):31-35, 2023. Second author.
4. Jianchao Chen, Zongbiao Weng, Zhiheng Liang, Yudong Li, Qiujie Chen, and Lei Wang. "A Shock-Absorbing and Self-Stabilizing Transport Platform for Fruits and Vegetables." Utility model patent.

RESEARCH AND INDUSTRY EXPERIENCE

Hong Kong University of Science and Technology (Guangzhou) | Guangzhou, China

Research Intern Feb 2025 - Jun 2025

- Developed a language-driven closed-loop semantic exploration pipeline integrating semantic perception, topological memory, subgoal decision making, local planning and obstacle avoidance, and learned locomotion control.
- Designed confidence-calibrated semantic evidence arbitration combining VLM scene reasoning with GroundingDINO detections, and built a controlled-growth semantic topological memory with utility-driven subgoal selection.
- Integrated high-level decisions with local planning and a reinforcement-learning locomotion policy; validated the system in NVIDIA Isaac Sim and on a Unitree Go1. The work led to the IROS 2026 paper above as a co-first author.

Zhiyuan Robotics (AgiBot) | Shenzhen, China

Perception and Planning Algorithm Intern Mar 2026 - May 2026

- Deployed ROS-based RGB-D, LiDAR, and ToF sensing modules in C/C++ and Python to build dense 3D point clouds and semantic environment representations for quadruped motion planning.
- Integrated object detection, semantic segmentation, DeepSORT, and Kalman filtering for dynamic-obstacle association, state estimation, and short-horizon motion-trend prediction used by local planning.
- Deployed and optimized HOV-SG/DOVSG semantic-navigation pipelines, linking perception, scene representation, task decision making, and execution; conducted validation and data collection in Isaac Sim and on a quadruped platform.

DJI | Shenzhen, China

Power Control Algorithm Intern Jul 2022 - Aug 2022

- Contributed to a 24 V/40 A bidirectional supercapacitor digital power system, including ADC acquisition and filtering, Buck-Boost dual-loop control, PWM regulation, protection logic, and board-level debugging.

SELECTED RESEARCH PROJECTS

Cross-Embodiment GNN-Diffusion Motion Planning with MPPI-MPC Recovery

Project Lead, SUSTech Sep 2024 - Present

- Developed a cross-embodiment execution pipeline connecting RGB-D/AnyGrasp target-pose estimation, morphology-conditioned joint-trajectory generation, and closed-loop terminal recovery.
- Parsed URDF models into link-joint-geometry graphs and used an edge-conditioned GIN to encode kinematic structure; conditioned a temporal diffusion model through morphology embeddings, AdaLN, target-pose cross-attention, and smooth trajectory interpolation.
- Integrated differentiable end-effector, self-collision, environment-collision, and path-length objectives, and implemented receding-horizon MPPI-MPC recovery for residual error, target motion, joint limits, and singular configurations.
- Evaluated five configurations in Isaac Sim and deployed on KD7/A10 robots: 74.3% mean zero-shot success; after adaptation, 91.7%/96.7% static-reaching and 86.7%/95.0% disturbance-recovery success. IEEE RA-L manuscript in preparation.

SELECTED RESEARCH PROJECTS (CONTINUED)

PPO-based Point-Foot Biped Locomotion Control

Project Lead, SUSTech Sep 2024 - Feb 2025

- Built parallel PPO training in Isaac Gym for commanded-velocity tracking with joint-position actions and low-level PD control; designed rewards for tracking, posture, gait, contact consistency, energy use, and action smoothness.
- Combined reward scheduling, termination design, and curriculum learning to stabilize gait formation, and randomized mass, friction, damping, and observations to improve robustness.
- Completed Isaac Gym-to-MuJoCo Sim-to-Sim evaluation and contributed to TRON2 Sim-to-Real deployment, system integration, and on-robot debugging.

Centroidal-Dynamics MPC for Bipedal Walking

Project Lead, South China Agricultural University Sep 2023 - Jun 2024

- Derived a 12-state three-dimensional centroidal model and formulated a constrained MPC-QP with a 10-step, 0.6 s horizon; used qpOASES to optimize ground-reaction forces and moments under friction and normal-force limits.
- Mapped optimized contact quantities through Jacobian-transpose control and combined Raibert foothold planning, swing-foot trajectories, inverse kinematics, and joint PD control in a hierarchical walking controller.
- Validated 0.5 m/s and 0.8 m/s command tracking and disturbance recovery in MATLAB/Simulink and MuJoCo; received the Excellent Undergraduate Thesis award.

Three-Axis Shock-Absorbing Stabilization Gimbal

Embedded Systems Lead May 2022 - Nov 2023

- Developed an STM32/FreeRTOS three-axis stabilization prototype, covering IMU acquisition, Kalman filtering, step-response modeling, system identification, and MATLAB/Simulink verification.
- Implemented and tuned an angle-outer-loop/angular-velocity-inner-loop cascaded PID controller through hardware-software integration and multi-condition disturbance testing.

RoboMaster Sentinel and Infantry Robot Embedded Control

Embedded Software Lead Aug 2021 - Aug 2022

- Developed STM32/FreeRTOS control software for sentinel and infantry robots, including gimbal cascaded PID control using IMU and encoder feedback, mecanum-chassis kinematics, and power management.
- Implemented CAN/UART/SPI/USB/Bluetooth communication, referee-system parsing, fault detection, system integration, and competition-site troubleshooting; contributed to national and regional RoboMaster awards.

TECHNICAL SKILLS

- Programming and frameworks: C/C++, Python, PyTorch, Linux, ROS/ROS2, MATLAB/Simulink.
- Planning and control: trajectory representation, interpolation, and generation; GNN/GIN, diffusion models, MPPI-MPC, MPC-QP, PPO, FK/IK, Jacobians, robot dynamics, PID control, and system identification.
- Perception and autonomy: RGB-D, LiDAR, ToF, 3D point clouds, semantic mapping, confidence calibration, GroundingDINO, SAM, YOLO, AnyGrasp, DeepSORT, and Kalman filtering.
- Simulation and deployment: NVIDIA Isaac Sim, Isaac Gym, MuJoCo, Webots; manipulator, biped, and quadruped simulation; Sim-to-Sim and Sim-to-Real deployment; STM32, FreeRTOS, CAN, UART, and SPI.

SELECTED AWARDS

- National First Prize, 8th International College Student Intelligent Agricultural Equipment Innovation Competition, 2023.
- Third Prize, Guangdong Regional Contest of the China Robot and Artificial Intelligence Competition, 2023.
- First Prize, Guangdong Engineering Student Comprehensive Experimental Skills Competition, 2022.
- National First Prize, RoboMaster 2022 Super Competition; National First Prize, RoboMaster 2022 Sentinel Robot Practical Combat Award; First Prize, RoboMaster 2022 Southern Regional Competition.
- Second-Class Scholarship, South China Agricultural University, 2022-2023; Third-Class Scholarship, 2020-2021 and 2021-2022.

LANGUAGE

Chinese (native); English: CET-4 558, CET-6 467.